MSDS 451 Assignment 02
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Problem Description

The objective of this study is to apply the Monte Carlo simulation methodology toward optimal portfolio allocation. Initially, four generic assets with specified expected return vectors, volatility profiles, and correlation matrices are evaluated to determine the optimal combination of long and short positions maximizing portfolio yield. Subsequently, the analysis evaluates four real-world equity assets using the same optimization engine to compare performance against the baseline generic asset framework. The broader goal is to assess the viability and efficacy of stochastic Monte Carlo techniques in constructing risk-optimized portfolios.

Data Preparation and Pipeline

Whereas the initial four generic assets were supplied with predetermined statistical parameters, empirical metrics were calculated for the second asset group. To evaluate portfolio resilience in the healthcare sector, four representative equities were selected: Eli Lilly (LLY), Johnson & Johnson (JNJ), UnitedHealth Group (UNH), and Intuitive Surgical (ISRG). These selections provide diverse exposure across risk, quality, and growth profiles. Ten years of historical monthly return data were extracted from Yahoo Finance to construct the expected return vector and covariance matrix. Target volatility thresholds were established based on historic standard deviations, reflecting distinct risk tolerance tiers.

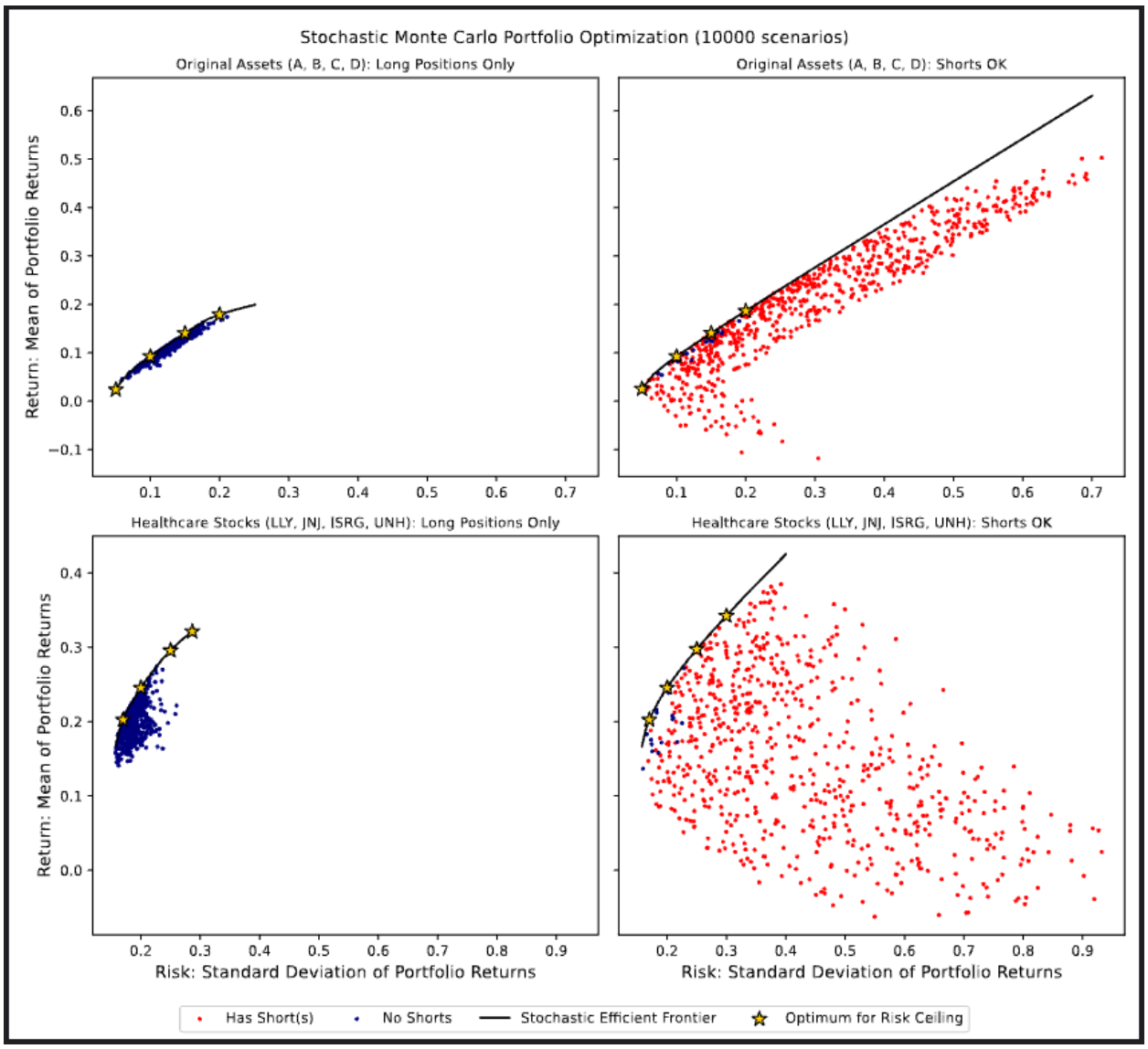
Research Design

The portfolio optimization framework employs a stochastic Monte Carlo simulation engine, modeling 10,000 potential future market scenarios across a one-year horizon—encompassing standard market conditions as well as extreme tail-risk events. The simulation matrix subjects candidate portfolios to rigorous stress testing. By evaluating performance across all generated scenarios, the optimization engine identifies the allocation that maximizes expected mean return subject to prescribed portfolio variance constraints.

This methodology expands upon conventional Monte Carlo frameworks, which often rely on randomized asset weights to map an efficient frontier based solely on historical expected returns. Integrating stochastic simulation across varied market regimes enables the optimizer to determine portfolio allocations that maintain robust performance during normal operations while mitigating downside exposure during stress events.

The simulation results are visualized by plotting all 10,000 scenarios, mapping standard deviation on the horizontal axis against expected return on the vertical axis. An efficient frontier curve illustrates the maximum return attainable for each risk level, highlighting optimal allocations under specific volatility limits. Key evaluation metrics include mean simulated return, risk-adjusted performance (Sharpe ratio), and downside risk (probability of negative return).

Results


Stochastic Monte Carlo Portfolio Optimization (10000 scenarios)
Original Assets (A, B, C, D): Long Positions Only
Original Assets (A, B, C, D): Shorts OK
Healthcare Stocks (LLY, JNJ, ISRG, UNH): Long Positions Only
Healthcare Stocks (LLY, JNJ, ISRG, UNH): Shorts OK
Return: Mean of Portfolio Returns
Risk: Standard Deviation of Portfolio Returns
Has Short(s)
No Shorts
Stochastic Efficient Frontier
Optimum for Risk Ceiling

Predefined Assets

Positions	Risk Ceiling	w1	w2	w3	w4	Simulated Return	Sharpe	Prob. Loss
Long Positions Only	Vol <= 5%	0.9252	0.0748	0.0000	0.0000	0.0237	0.4742	0.3192
Long Positions Only	Vol <= 10%	0.4951	0.0000	0.3559	0.1490	0.0922	0.9223	0.1753
Long Positions Only	Vol <= 15%	0.1676	0.0000	0.5697	0.2627	0.1400	0.9330	0.1768
Long Positions Only	Vol <= 20%	0.0000	0.0000	0.3969	0.6031	0.1788	0.8940	0.1865
Shorts OK	Vol <= 5%	0.8925	0.0946	0.0369	-0.0240	0.0251	0.5023	0.3075
Shorts OK	Vol <= 10%	0.5147	-0.0299	0.3621	0.1531	0.0923	0.9226	0.1756
Shorts OK	Vol <= 15%	0.2443	-0.1190	0.5950	0.2797	0.1403	0.9355	0.1759
Shorts OK	Vol <= 20%	-0.0141	-0.2040	0.8175	0.4007	0.1862	0.9312	0.1790

Healthcare Stocks

Positions	Risk Ceiling	LLY	JNJ	ISRG	UNH	Simulated Return	Sharpe	Prob. Loss
Long Positions Only	Vol <= 17%	0.3317	0.4032	0.1055	0.1596	0.2024	1.1909	0.1158
Long Positions Only	Vol <= 20%	0.5358	0.1722	0.1310	0.1609	0.2454	1.2271	0.1059
Long Positions Only	Vol <= 25%	0.8079	0.0000	0.1179	0.0742	0.2956	1.1825	0.1156
Long Positions Only	Vol <= 30%	1.0000	0.0000	0.0000	0.0000	0.3212	1.1210	0.1304
Shorts OK	Vol <= 17%	0.3317	0.4032	0.1055	0.1596	0.2024	1.1909	0.1158

Positions	Risk Ceiling	LLY	JNJ	ISRG	UNH	Simulated Return	Sharpe	Prob. Loss
Shorts OK	Vol <= 20%	0.5358	0.1722	0.1310	0.1609	0.2454	1.2271	0.1059
Shorts OK	Vol <= 25%	0.7826	-0.1050	0.1632	0.1592	0.2974	1.1895	0.1116
Shorts OK	Vol <= 30%	0.9962	-0.3458	0.1908	0.1589	0.3424	1.1412	0.1240

Conclusion

Comparing the generic asset set against the healthcare stock portfolio demonstrates noticeable structural differences in simulation outputs. The generic assets exhibit a linear relationship between expected return and volatility. While the healthcare portfolio follows a similar directional trend, it shows greater dispersion in expected returns across equivalent volatility levels. Notably, the efficient frontier for the healthcare assets displays a significantly steeper trajectory, indicating superior return potential at lower relative risk thresholds.

Two distinct optimization constraints were evaluated: a long-only strategy and an unconstrained strategy permitting short positions. Under a 30% volatility ceiling, the long-only model concentrates 100% of the portfolio into LLY to maximize growth potential. When short positions are permitted, shorting JNJ to finance additional allocation in LLY increases the mean expected return to 34%, with a 12.4% probability of loss. Alternatively, for risk-averse objectives, a 20% volatility cap achieves an optimal balance between Sharpe ratio maximization and downside protection. Under both constrained and unconstrained conditions, the 20% target allocates predominantly to LLY with minor weights in the remaining assets, producing a mean return of ~24.5% and a ~10% probability of loss.

Stochastic Monte Carlo simulation offers a robust and adaptable framework for asset allocation. By stress-testing candidate portfolios against comprehensive scenario distributions, the methodology produces well-supported asset weighting strategies. Optimizing across diverse economic regimes enhances portfolio stability and consistency across changing market environments.